PIXEL TECH.

BRD

Web-based Electronic Gate Pass System

Prepared For

MAJIS

Version: 1.0

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1. Introduction

Majis Industrial Services S.A.O.C seeks to implement a modern, secure, and fully integrated Electronic Gate Pass System to streamline the process of issuing access permits for visitors, contractors, employees, and vehicles entering Sohar Port. This initiative aims to support operational efficiency, digital transformation, and compliance with Sohar Port and Royal Oman Police (ROP) access control standards.

The Pixel Tech Team offers a solution by developing a web app that allows visitors to submit requests via a seamless web form. Additionally, an Admin portal will be established for the Majis Team to manage these requests, providing the ability to approve or reject them. This portal will integrate with the Sohar Port access control system through APIs, enabling the creation of PDF files containing QR codes for entry registration.

We created this document to compile all User Stories available on the Web App, ensuring clarity on every aspect of the App. The Majis Team will examine the details and give their approval.

2. Project Objectives

- **Core Objective:** The main goal is to build a **secure, web-based platform** that enables visitors, contractors, employees, and vehicle owners to apply for gate passes online.
- **Internal Control:** The system must provide Majis with an **internal approval workflow** for managing and reviewing these applications.
- **External Integration:** A critical objective is to ensure **seamless integration** with the **Sohar Port access control system through APIs**.

3. Scope Of Work

- **Web-Based Platform:** The system will be delivered as a web-based, bilingual platform (Arabic & English) hosted under Majis' domain.
- **User Application Portal:** A public-facing, user-friendly interface for visitors, contractors, employees, and drivers to submit gate pass applications online.
- **Admin Portal & Workflow Management** that include a dashboard with summary metrics, detailed request views, Approval/Rejection functionality with notes, search/filter tool.

- **API Integration with Sohar Port:** This is a core component involving REST-based integration using JSON payloads, API Key authentication, and HTTPS encryption. The integration will handle **1 GET API** and **2 POST APIs** to submit approved data and validate responses (approval/decline codes).

4. Functional Requirements

4.1 Visitors Portal

ID	vis_01
Title	Visitors Portal
Description	Individuals visiting the Majis company site are required to have a Permit issued for entry at the company entrance. Through this portal, users can access a streamlined form containing several fields that Visitors need to complete. This portal provides a user-friendly, public-facing web form for all users to submit their access requests to the Admin Team for approval or rejection.
Inputs/ Views	The submission form must include, but is not limited to, the following fields: ● Name ● Email ● Passport/ID Number (image) ● Purpose of visit ● Date of Visit (Date) ● Request Type (Dropdown/Selection for Visitor, Contractor, Employee, Vehicle). ● ETC … (More data will come from The Majis Team)
Preconditions	**No login or sign-up is required** to access and submit the form. The system must be available and connected to the database.
Postconditions	● **Successful Submission(alert):** A **successful alert** message appears in the app confirming submission. ● **Successful Submission(Visitor):** A **Notification Email** is automatically sent to the Visitor with a confirmation message

	from our system. (The Majis Team will provide the content format). ● **Successful Submission(Admin):** A **Notification Email** is automatically sent to **the Admin(MAJIS Team)** upon updating a new request in the system. ● **Validation Failure:** An **error alert** is displayed if mandatory fields are not completed, preventing submission.
Steps	1- The visitors will use the Visit Portal URL to reach the form 2- The Page **opens without any restrictions** (no authentication required). 3- The user **views all input fields** required for their request type. 4- The user **fills all mandatory fields**. 5- The user **clicks 'Submit'** only when all mandatory fields are completed
Test Cases	**Case 1:** Successful Submission: ● **Action:** The user correctly fills all mandatory fields and submits the form. ● **Expected Result:** A successful alert appears. The request is successfully stored in the database (status: Pending). The system sends a confirmation email to the visitor. **Case 2:** Mandatory Field Validation: ● **Action:** The user leaves the "Passport/ID Number" field empty and attempts to submit. ● **Expected Result:** An error alert is created, and the form is not submitted. A clear message indicates which fields are missing. **Case 3:** Email Notification Trigger: ● **Action:** Submit a valid request. ● **Expected Result:** The system immediately sends a notification email to the Email address provided in the form.

4.2 Admin Portal

4.2.1 Login

ID	ad_01
Title	Login Page
Description	The admin will have a credential login by email and Password to enroll to the Dashboard.
Inputs/ Views	The inputs are: ● Email ● Password ● Insert OTP
Preconditions	● The system must be connected to the administrator credentials database. ● All fields must be filled before submission is allowed (Cannot login without filling all fields).
Postconditions	● **Success and Redirect:** Redirect to the main page (Dashboard) in the Admin Portal after login is successful. ● **Success:** A successful alert is displayed once the submission and data are correct. ● **Failure:** An error alert is created once the credential login is wrong.
Steps	● Go to the login page URL ● Write your Email and Pass. ● Click the "Login" button.
Test Cases	Case 1: Successful Login: ● **Action:** Enter correct Email and Password. ● **Expected Result:** A successful alert is displayed, and the system redirects to the Main Page in the Admin Portal . Case 2: Invalid Credentials:

	• **Action:** Enter incorrect Password. • **Expected Result:** An error alert is created, and the user remains on the Login Page . **Case 3: Missing Fields:** • **Action:** Leave the Email field empty and attempt to log in. • **Expected Result:** A validation message/alert is displayed, preventing submission (Cannot login without filling all fields).

4.2.2 Statistics and KPIs

ID	St_01
Title	Statistics and KPIs
Description	The admin will have an overview of system performance through statistics and KPIs. This function allows administrators to gain a comprehensive overview of the system's performance and workload through the display of up-to-date statistics and KPIs on an interactive dashboard.
Inputs/ Views	**Dashboard View displaying the following:** 1- Total requests of Visitors. 2- Total approved requests. 3- Total rejected requests. 4- Total pending requests. 5- **Line charts** created to track the **time-based trend** of approved and rejected requests. 6- Design a **pie chart** to illustrate the comparison of approved, rejected, and pending requests.
Preconditions	Must be login by Admin (email && Password) to see the results.
Postconditions	The Dashboard is fully displayed and loaded with updated statistical data and indicators immediately upon the administrator accessing the page.
Steps	1- The administrator **logs in** to the "Admin Portal" successfully. 2- The administrator is **redirected** to the main Dashboard.

	3- The system **displays** all key metrics and counters (Total, Approved, Rejected, Pending). 4- The system **displays** the updated Line and Pie charts.
Test Cases	**Case 1:** Data Consistency Check: ● **Action:** Sum the counts of approved, rejected, and pending requests. ● **Expected Result:** The sum of these counts must exactly match the Total requests of Visitors counter. **Case 2:** Pie Chart Display Accuracy: ● **Action:** View the Pie chart. ● **Expected Result:** The chart must accurately represent the percentage distribution of request statuses (Approved, Rejected, Pending), summing up to 100%.

4.2.3 View and Review Requests

ID	vr_01
Title	View and Review Requests
Description	This function allows administrators on the Majis Team to view all gate pass requests submitted via the "Visitors Portal" examine the details of each request, and proceed with the approval or rejection process.
Inputs/ Views	A table displaying a summary of each request, including: ● Request ID ● Applicant Name ● Date Submitted ● Purpose of Visit ● Status (Pending, Approved, Rejected) ● Date of Visit ● Request Type (Visitor, Contractor, Employee, Vehicle) ● Request action (Approved, Rejected). ● Request Details View: it will be A separate page showing all data entered by the visitor (Name, Email, Passport, etc.), along with the review log and notes. ● API Response: - Receive API code status after approved (Number)

	- Receive API message after approved (if applicable) (String)
Preconditions	The administrator **must be successfully logged in** to the "Admin Portal."
Postconditions	**Upon accessing the list:** An updated list of all requests is displayed, prioritizing pending requests, showing the latest available data.
Steps	1- **Navigate** to the main Dashboard in the "Admin Portal." 2- **Click** on the "Requests Management" or "Requests" tab. 3- **View** the table containing the list of requests. 4- **Use** the search and filter tools (Search/Filter tool) to narrow down the list (by status, date, or user type (Request Type)). 5- **Click** on a specific request (Request ID) to open the **Details View**. 6- **Review** the entered data and any attachments (if applicable). 7- **Take action:** Select "Approve" or "Reject," entering notes (Notes) if necessary.
Test Cases	**Case 1:** Filtering Pending Requests: ● **Action:** The administrator applies a filter for the "Pending" status. ● **Expected Result:** The list displays all requests that have not yet been acted upon. **Case 2:** Successful Detail View: ● **Action:** The administrator clicks on a valid request ID. ● **Expected Result:** The details page opens, and all data fields entered by the visitor are accurately displayed.

4.2.4 Editing Requests with Integration Sohar Port

ID	ed_01
Title	Editing Requests with Sohar Port System Integration
Description	This function allows the administrator to review a gate pass request, **edit any data entered by the visitor** if necessary, and then approve the request and

	automatically submit it via the API interface to the Sohar Port access control system.
Inputs/ Views	● **Details View:** All visitor data fields (Name, Passport, Date, etc.) are displayed as **Editable Fields** for the administrator. ● An **Approve** button and a **Reject** button. ● A mandatory field for **Rejection Notes**.
Preconditions	● The administrator **must be successfully logged in**. ● The request must be in the **"Pending"** status. ● The Sohar Port integration data (API Key) **must be valid**.
Postconditions	**In Case of Approval (Approve):** ● Any edits made by the administrator to the visitor's data are **saved**. ● The **POST API interface** of the Sohar Port system is called to send the approved data. ● The request status is updated to **"Approved"**. ● The request with Data sent to **Sohar port** by API, the final approve will be by **Sohar Port Management System**. ● API Response of Sohar Port Integration: - Receive API code status after approved (Number). - Receive API message after approved (if applicable) (String). ● A **PDF file** containing a **QR code** for entry registration is **generated (by Sohar Port Automatically)**. ● An **email notification** is sent to the visitor including the PDF file **generated by (Sohar Port Automatically)**. **In Case of Rejection (Reject):** ● The request status is updated to **"Rejected"**. ● An **email notification** is sent to the visitor including the rejection message and the entered notes **generated by (Majis Team and the system will send email automatically**).
Steps	1- The administrator **opens** the details page for a request in "Pending" status

	2- The administrator **reviews** the data and **edits** any field that requires correction (e.g., change/add a vehicle number). 3- The administrator **clicks** the **"Approve"** button. 4- The system **checks** for successful submission via API to Sohar Port. 5- **If submission is successful:** The status changes to "Approved," the QR file is generated when approved by Sohar port system, and the email is sent. 6- **If submission fails:** An **Error Alert** is displayed, and the request remains in "Pending" status with a failure note. 7- <u>(Alternatively, to Step 3)</u> The administrator **clicks** the **"Reject"** button and enters the mandatory rejection notes. 8- The administrator **confirms** the rejection, the status changes to "Rejected," and the email is sent.
Test Cases	**Case 1:** Approval with Editing and Successful Integration: ● **Action:** The administrator edits the "Purpose of Visit" field and approves the request. ● **Expected Result:** The edit is saved, the POST API call is successful, the status changes to "Approved" after Sohar port make final approve the visitor receives the email with the QR file. **Case 2:** Rejection without Editing: ● **Action:** The administrator clicks "Reject" and enters the rejection reason (e.g., "Missing ID documentation"). ● **Expected Result:** The status changes to "Rejected," and the rejection email is sent with the notes. **Case 3:** Integration Failure (API Integration Failure) ● **Action:** The administrator approves a request, but the integration with Sohar Port fails (the system receives an error code from the API). ● **Expected Result:** An error message is displayed to the administrator, and the request status remains **"Pending"** to prevent false approvals.

4.2.5 Rules and Permissions

ID	rp_01
Title	Role and Permission Management for Administrators
Description	This function ensures that strict access rules are applied to administrators' permissions based on their assigned role **(Super Admin or Sub Admin)**, guaranteeing that each user can only access the tasks necessary for their function.
Inputs/ Views	**Admin Portal:** Login and User Management interfaces that have Suber Admin and Sub Admin**.** **Super Admin has access to all portal with:** ● An interface to add/edit/delete **Sub Admin accounts** (Name & Email & Password). ● An interface to define specific permissions for each **Sub Admin** access or not to: - View Dashboard & Statistics - View Requests List - Logs - User Management (Create another Admins, make Permissions ETC…) Note: **The Pixel Technical Team will set up a single Super Admin account and inform specific members of the MAJIS Team, granting them complete access to the Admin Portal and the ability to add Sub Admins.**
Preconditions	The system must authenticate and assign the user as either a "Super Admin" or "Sub Admin" upon successful login.
Postconditions	The administrator's main interface must be loaded according to their assigned permissions; unauthorized functions must be hidden or blocked.
Steps	1- The administrator **logs in** successfully. 2- The system **verifies** the user's role. 3- **If "Super Admin":** All system functionalities, including User Management, are displayed.

	4- **If "Sub Admin":** is displaying only the access permission provided by the Super Admin.
Test Cases	**Case 1:** Sub Admin Access to Statistics: ● **Action:** A "Sub Admin" logs in and attempts to access the Statistics Dashboard. ● **Expected Result:** Access should be granted successfully (consistent with their primary function of reviewing requests). **Case 2:** Blocking Sub Admin from User Management: ● **Action:** A "Sub Admin" logs in and attempts to access the "User Management/Permissions" page. ● **Expected Result:** Access is denied, and an "Unauthorized Access" error message is displayed. **Case 3:** Super Admin's Ability to Assign Permissions: ● **Action:** A "Super Admin" adds a new "Sub Admin" account and assigns "Approve and Reject" permissions. ● **Expected Result:** The account is created successfully, and upon logging in, the new "Sub Admin" can Approve or Reject requests.

4.2.6 Activity Logs Page

ID	lg_01
Title	Activity and Audit Logs
Description	This function allows the Super Admin to review a comprehensive record of all critical actions performed within the system, including logins, request approvals, data edits, **and all integration interactions with the Sohar Port API**, ensuring auditing and accountability.
Inputs/ Views	**Logs Table:** The table must display the following information for each event: ● **Date & Time:** The precise time the action occurred. ● **User ID/Name:** The identity of the administrator who performed the action or the system (for automated actions). ● **Action Type:** - Request Management: all actions related with visitors requests flows - User Management: Create/Delete/Update/ Permissions for Sub Admin users. - System Integration: Any reactions with API Sohar Port [Sent: POST, Status: Success, Status: Failure] **API NOTE:** If there are no API requests, we will display an empty message saying "No requests created." Or ● **Action Performed:** A specific description of the action (e.g., "Login Success," "Approved Request #123," "Sub Admin Account Created"). ● **Affected Entity Type:** The type of record affected (Request, User, API Call). ● **Filter & Search Tools:** Must be available for searching by Date, Action Type, User ID/Name.
Preconditions	Access to this page must be restricted to **Super Admin** only.
Postconditions	The latest activity logs are loaded and displayed in the Logs Table in reverse chronological order (newest first).

Steps	1- The Super Admin **logs in** to the "Admin Portal." 2- The Super Admin **navigates** to the "Activity Logs" page. 3- The system **displays** the logs table, sorted from newest to oldest. 4- The Super Admin **uses** the search and filter tools to investigate a specific action or time period.
Test Cases	**Case 1:** Successful API Integration Logging: ● **Action:** A Sub Admin approves a request, and the API call succeeds. ● **Expected Result:** A new log entry appears showing: "the admin's name, the action "Approval of Request #123," and Response Code 200 logged in the API Data field." **Case 2:** API Integration Failure Logging: ● **Action:** An Admin approves a request, but the API integration fails (e.g., due to an invalid API Key). ● **Expected Result:** A new log entry appears showing: "the admin's name, the action "Attempted API Submission for Request #124," and Response Code (e.g., 401) and the error message received from the Sohar Port API are logged." **Case 3:** Logging Administrative Action: ● **Action:** The Super Admin changes the permissions of a Sub Admin. ● **Expected Result:** A new log entry appears showing the action "Updated Permissions for Sub Admin [Name]".

This section to know more about the log's types
(Example)

Action Type	Action Performed	Affected Entity Type
Request Management	Approved Request #[Request ID]	Request
Request Management	Rejected Request #[Request ID]	Request
Request Management	Edited Visitor Data on Request #[Request ID] (e.g., Name, Date, Passport)	Request
Request Management	Viewed Request Details (for high-security auditing)	Request
User Management	Sub Admin Account Created: [User Name]	User
User Management	Sub Admin Account Updated (e.g., Password, Email)	User
User Management	Updated Permissions for User: [User Name]	User
System Integration	API Call Sent: POST to Sohar Port for Request #[Request ID]	API Call
System Integration	API Call Status: Success (Code 200) for Request #[Request ID]	API Call
System Integration	API Call Status: Failure (Code 401/500). Error: [Error Message from API]	API Call